# Wave guides as an analogous system of 1D optical lattice for anyons

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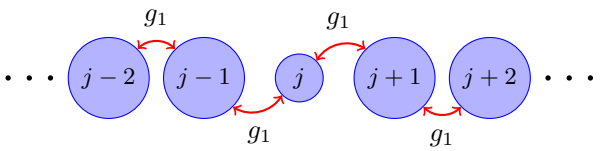
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Keywords: anyons, quantum matter, ultracold atoms

## I. INTRODUCTION

Los anyones son ...

## II. MAIN IDEA




Para las guías de onda:

- Una guía de onda pose un indice de refracción en el otro núcleo, rodeado por otro de menor indice $n_1 < n_2$
- Para una única guía de onda, los modos admitidos están dados por sus características físicas.
- Un arreglo de guías de onda, el acoplamiento permitirá la transferencia entre los elementos del arreglo.

From an infinite waveguides array with homogeneous optical single mode waveguides and equally spaced along z-axis. This is described by the following differential equation [1]

$$i\frac{dE_j(z)}{dz} = g_1(E_{j-1} + E_{j+1}), \quad j = -\infty, \ldots, \infty, \quad (1)$$

From this

$$i\frac{d|\psi(z)\rangle}{dz} = \hat{H}|\psi(z)\rangle, \quad \hat{H} = g_1\left(\hat{V}_\infty + \hat{V}_\infty^\dagger\right), \quad (2)$$

where the operators $\hat{V}_\infty$ and $\hat{V}_\infty^\dagger$ are the infinite counterparts of the London-Susskind–Glogower operators [2, 3], defined as

$$\hat{V}_\infty = \sum_{n=-\infty}^{\infty} |n\rangle\langle n+1|, \quad \hat{V}_\infty^\dagger = \sum_{n=-\infty}^{\infty} |n+1\rangle\langle n|, \quad (3)$$

For an analytical solution we need to satisfy in our Hilbert space the following conditions for the LSG operator

$$\left[\hat{V}_\infty, \hat{V}_\infty^\dagger\right] = 0, \qquad \hat{V}_\infty\hat{V}_\infty^\dagger = \hat{I},$$

In this framework the initial state is a photon in the $n$ site of an infinite array of waveguides $|\psi(0)\rangle = |n_0\rangle$,:

$$|\psi_{n_0}(z)\rangle = \exp\left[-g_1 z\left(i\hat{V}_\infty^\dagger - \frac{1}{i\hat{V}_\infty^\dagger}\right)\right]|n_0\rangle\,.$$

Donde podemos identificar la función generadora de las funciones Bessel, para obtener

$$|\psi_{n_0}(z)\rangle = \sum_{n=-\infty}^{\infty} J_n(-2g_1 z) i^n \hat{V}_\infty^{\dagger n}|n_0\rangle\,.$$

para este modelo podemos hacer una analogía con un modelo de bosones en una red óptica partiendo del operador $\hat{V}_\infty$ y la evolución será escrita en función del tiempo:

$$\hat{V}_\infty = \sum_{j=-\infty}^{\infty} \hat{b}_j^\dagger \hat{b}_{j+1}, \quad \hat{V}_\infty^\dagger = \sum_{j=-\infty}^{\infty} \hat{b}_{j+1}^\dagger \hat{b}_j, \quad (4)$$

Rewriting the Hamiltonian as

$$\hat{H} = -J\sum_{j=-\infty}^{\infty}\left(\hat{b}_j^\dagger\hat{b}_{j+1} + \hat{b}_{j+1}^\dagger\hat{b}_j\right) \quad (5)$$

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The operator $\hat{V}_{\infty}^{\dagger n}$ is equivalent of create a particle in $n$ site in the framework of optical lattices for a subspace of one particle per site:

$$\hat{V}_{\infty}^{\dagger n}|1_0\rangle = \hat{b}_n^{\dagger}|0\rangle, \tag{6}$$

Obtaining the dynamics for a particle in 0-site

$$|\psi_0(t)\rangle = \sum_{n=-\infty}^{\infty} J_n(-2Jt) i^n \hat{b}_n^{\dagger}\,|0\rangle\,. \tag{7}$$

We could obtain a rich dynamics for n-particles system working with a Heisenberg representation for one body operators $\hat{b}^{\dagger}$ applied in the sites, obtaining in general the expression:

$$\langle \hat{n}_j \rangle = \sum_{m=-\infty}^{\infty} |J_{m-j}(-2Jt)|^2 \langle \hat{n}_m(0)\rangle \tag{8}$$

$$\langle \mathbf{J}_j \rangle = iJ \left\langle \hat{b}_{j+1}^{\dagger}\hat{b}_j - \hat{b}_j^{\dagger}\hat{b}_{j+1} \right\rangle \tag{9}$$

$$\langle \mathbf{T}_j \rangle = \sum_{m=-\infty}^{\infty} |J_{m-j}(-2Jt)|^2 \langle \hat{n}_m(0)\rangle \tag{10}$$

also:

$$\mathbf{J} = iJ \sum_{i=1}^{M} \left( e^{-i\theta\hat{n}_i}\hat{b}_{i+1}^{\dagger}\hat{b}_i - \hat{b}_i^{\dagger}\hat{b}_{i+1}e^{i\theta\hat{n}_i} \right) \tag{11}$$

$$\mathbf{T} = J \sum_{i=1}^{M} \left( e^{-i\theta\hat{n}_i}\hat{b}_{i+1}^{\dagger}\hat{b}_i + \hat{b}_i^{\dagger}\hat{b}_{i+1}e^{i\theta\hat{n}_i} \right) \tag{12}$$

$$\mathrm{S}(q) = \sum_{l=0}^{M} |\mathrm{J}_l(-2Jt)|^2 - \sum_{l,m=0}^{M} e^{\mathrm{iq}(l-m)} |\mathrm{J}_l(-2Jt)|^2 |\mathrm{J}_m(-2Jt)|^2$$

## III. CONCLUSION


[1] Alan L. Jones. Coupling of optical fibers and scattering in fibers. *J. Opt. Soc. Am.*, 55:261–271, Mar 1965.

[2] Fritz London. Über die jacobischen transformationen der quantenmechanik. *Z. Phys.*, 37:915–925, Dec 1926.

[3] Leonard Susskind and Jonathan Glogower. Quantum mechanical phase and time operator. *Physics Physique Fizika*, 1:49–61, Jul 1964.


### Appendix: Appendix

In this paper we suggest to calculate the operator from active representations due the feasibility of operations

#### 1. Passive representation of $\hat{n}_j$ operator

For this framework we need to specify the initial state, first, for 2 particles in the sites $j$, $j'$ respectively, with $j \neq j'$, the state as function of time is

$$|\psi(t)\rangle = e^{-i\mathcal{H}t}|\psi(0)\rangle = e^{-i\mathcal{H}t}\hat{b}_j^{\dagger}\hat{b}_{j'}^{\dagger}|0\rangle, \tag{A.1}$$

as the temporal propagator is unitary we use the identity to get the following expressions

$$\begin{aligned} |\psi(t)\rangle &= e^{-i\mathcal{H}t}\hat{b}_j^{\dagger}\mathcal{I}\hat{b}_{j'}^{\dagger}\mathcal{I}|0\rangle \\ &= e^{-i\mathcal{H}t}\hat{b}_j^{\dagger}e^{i\mathcal{H}t}e^{-i\mathcal{H}t}\hat{b}_{j'}^{\dagger}e^{i\mathcal{H}t}e^{-i\mathcal{H}t}|0\rangle \\ &= e^{-i\mathcal{H}t}\hat{b}_j^{\dagger}e^{i\mathcal{H}t}e^{-i\mathcal{H}t}\hat{b}_{j'}^{\dagger}e^{i\mathcal{H}t}|0\rangle, \end{aligned} \tag{A.2}$$

as the bosons obey time invariance $\hat{U}^{\dagger}\hat{b}^{\dagger}\hat{U} = \hat{U}\hat{b}^{\dagger}\hat{U}^{\dagger}$

$$|\psi(t)\rangle = \hat{b}_j^{\dagger}\hat{b}_{j'}^{\dagger}|0\rangle, \tag{A.3}$$

we derive the Heisenberg equation of motion $\hat{b}^{\dagger}(t)$ from

$$i\frac{d}{dt}\hat{b}_j^{\dagger} = \left[\hat{b}_j^{\dagger}, \hat{H}\right], \tag{A.4}$$

doing the commutator with (5) obtain:

$$i\frac{d}{dt}\hat{b}_j^{\dagger}(t) = J\left(\hat{b}_{j-1}^{\dagger} + \hat{b}_{j+1}^{\dagger}\right), \tag{A.5}$$

consistent with the phenomenology of (1), to solve this differential equation we can use the Fourier transform $\hat{b}_j^{\dagger} = \frac{1}{\sqrt{N}}\sum_k e^{-ikj}\hat{b}_k^{\dagger}$ and return to the original space with an inverse Fourier transform, so in momentum space the differential equation decouples:

$$i\frac{d}{dt}\hat{b}_k^{\dagger}(t) = 2J\cos(k)\hat{b}_k^{\dagger}(t), \tag{A.6}$$

Solving

$$\hat{b}_k^\dagger(t) = e^{-i2Jt\cos(k)}\hat{b}_k^\dagger(0), \tag{A.7}$$

and taking the continuum limit to transform back to time space using the inverse Fourier transform

$$\hat{b}_j^\dagger(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-ikj}\left(e^{-i2Jt\cos(k)}\hat{b}_k^\dagger(0)\right)dk, \tag{A.8}$$

substituting $\hat{b}_k^\dagger(0) = \sum_m e^{ikm}\hat{b}_m^\dagger(0)$

$$\hat{b}_j^\dagger(t) = \sum_m\left(\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-i2Jt\cos(k)}e^{ik(m-j)}\hat{b}_m^\dagger(0)dk\right), \tag{A.9}$$

using the Jacobi Anger expansion:

$$e^{iz\cos(k)} = \sum_{n=-\infty}^{\infty} i^n J_n(z)e^{ink}. \tag{A.10}$$

Substituting with $z = -2Jt$, we obtain

$$\hat{b}_j^\dagger(t) = \sum_{m,n}\left(\frac{1}{2\pi}\int_{-\pi}^{\pi} i^n J_n(-2Jt)e^{ik(m+n-j)}\hat{b}_m^\dagger(0)dk\right), \tag{A.11}$$

due orthogonality of complex exponentials functions, the integral will be zero unless $n+m-j=0$, so for the values $n = j-m$ and using properties of Bessel function, we get obtain the the expression:

$$\hat{b}_j^\dagger(t) = \sum_{m=-\infty}^{\infty} i^{m-j}J_{m-j}(-2Jt)\hat{b}_m^\dagger(0) \tag{A.12}$$

### 2. Active representation of $\hat{n}_j$ operator

We obtain for a $|\psi\rangle$ as steady state and $\hat{b}^\dagger$ as dynamic operator, write as:

$$\hat{b}_j^\dagger(t) = \sum_{m=-\infty}^{\infty} i^{m-j}J_{m-j}(-2Jt)\hat{b}_m^\dagger(0), \tag{A.13}$$

so the expected value of $\hat{n}_j$ is:

$$\begin{aligned}\langle\hat{n}_j(t)\rangle &= \langle\psi(0)|\hat{b}_j^\dagger(t)\hat{b}_j(t)|\psi(0)\rangle,\\ &= \langle\psi(0)|\sum_{m,m'=-\infty}^{\infty} i^{m-m'}J_{m-j}(-2Jt)J_{m'-j}(-2Jt)\\ &\quad \hat{b}_m^\dagger\hat{b}_m|\psi(0)\rangle,\end{aligned} \tag{A.14}$$

In general we get a 1-d chain, if we annihilate at $m'$ site and create at $m$ we will obtain an orthogonal state, so for the different values of $m \neq m'$ we will obtain zero, so for an arbitrary initial state we get:

$$\langle\hat{n}_j(t)\rangle = \sum_{m=-\infty}^{\infty}|J_{m-j}(-2Jt)|^2\langle\hat{n}_m(0)\rangle \tag{A.15}$$

in the case of anyons using the equation (5), .... reducing for $N=2$

$$i\frac{d}{dt}\hat{b}_j(t)) = -J\left(\hat{b}_{j-1}+\hat{b}_{j+1}\right) - J\left(\left(e^{-i\theta}-1\right)\hat{n}_{j-1}\hat{b}_{j-1} + 2\left(e^{i\theta}- \right.\right. \tag{A.16}$$

$$\begin{aligned}i\frac{d}{dt}\hat{b}_j(t) = &-J\left(\hat{b}_{j-1}+\hat{b}_{j+1}\right)\\ &- J\Big[\left(e^{-i\theta}-1\right)\hat{n}_{j-1}\hat{b}_{j-1}\\ &+ 2\left(e^{i\theta}-1\right)\hat{n}_j\hat{b}_{j+1}\\ &+ \left(e^{-i\theta}-1\right)\hat{b}_{j+1}\hat{b}_j^2\Big]\end{aligned} \tag{A.17}$$